

Structural Fidelity / Use-Validity Boundary

***Representation Similarity, Required Structure, and
the Limits of Downstream Use***

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Abstract

This document formalizes the **Structural Fidelity / Use-Validity Boundary** as a conceptual boundary for distinguishing resemblance to a source from adequacy for a downstream task. A synthetic, simulated, transformed, compressed, or proxy representation may preserve visible features of a source system while failing to preserve the relation required for inference, comparison, prediction, intervention, evaluation, or decision.

The central distinction is that structural fidelity is always fidelity to something. A representation may reproduce distributions, categories, frequencies, correlations, layouts, output forms, or recognizable patterns. These properties can be useful and accurately measured. Yet the intended use may depend on ordering, dependency, causal sequence, subgroup variation, temporal constraint, role hierarchy, traceability, or another relation not captured by the fidelity measure. In that case, a representation can be faithful at one layer and invalid at the layer where it is used.

The note does not reject synthetic data, simulations, summaries, models, or proxies. It requires that the preserved structure and the use-required structure be declared separately. The framework is relevant wherever similarity metrics, realistic surfaces, or source-like outputs create pressure to expand use beyond the conditions actually tested. It provides a vocabulary for use-specific validation, metric placement, proxy-use expansion, and the difference between access restoration and evidential sufficiency.

Keywords

Structural Fidelity; Use-Validity; Representation Boundary; Downstream Validity; Fidelity–Validity Separation; Synthetic Representation; Proxy Data; Mechanism Preservation; Ordered Dependency; Inferential Adequacy; Validation Scope; Use Fitness; Metric Misplacement; Meta-Writing Ecology.

1. Introduction

Representations are often evaluated by how well they resemble what they represent. This comparison is necessary, but it can become structurally incomplete when the representation is later used for a purpose that demands a different relation. A dataset can reproduce marginal distributions while losing dependencies. A summary can preserve main claims while removing qualifications needed for policy use. A simulation can generate plausible outcomes without preserving intervention logic. A proxy can restore access while remaining unsuitable for adjudication.

The resulting error is not simply inaccuracy. The representation may be accurate according to the metric chosen. The error appears when evidence of resemblance is converted into evidence of fitness for a different task.

This boundary note therefore separates two questions: What structure has been preserved? What structure does the intended use require? The relation between those answers determines use-validity. The framework is designed to prevent a general fidelity claim from silently becoming a broad license for downstream use.

2. Core Definition

Structural Fidelity / Use-Validity Boundary describes the condition in which a synthetic, generated, simulated, transformed, compressed, or proxy representation preserves visible structural features of a source system while failing to preserve the structure required for a specific downstream use.

The representation may appear faithful. It may reproduce distributions, categories, frequencies, correlations, surface patterns, temporal ranges, or recognizable output forms. It may look sufficiently similar to the source system under general comparison metrics.

However, the intended use may depend on deeper relations that are not captured by surface fidelity alone.

These deeper relations may include ordering, dependency, causal sequence, intervention structure, role hierarchy, conditional mechanism, evaluation pathway, traceability, subgroup variation, temporal constraint, or decision-relevant relation.

The boundary problem appears when fidelity to visible structure is treated as validity for use.

In this condition, the system is not necessarily inaccurate at the level it was optimized to reproduce. The failure emerges because the preserved structure is not the structure required by the use case.

3. Classification and Scope

This document presents **Structural Fidelity / Use-Validity Boundary** as an external-facing boundary or diagnostic note. The note isolates a recurring distinction

that supports several broader models without claiming an independent universal mechanism in every domain.

It is not a legal, ethical, clinical, technical, or compliance standard. It does not assign blame, calculate risk, or authorize automated decisions. The public version omits non-public editorial records, private source materials, non-public working materials, and implementation details.

4. Central Distinctions

structural fidelity $\neq$ use-validity

Expanded:

A representation may preserve the visible structure of a source system without preserving the structure required for the inference, decision, comparison, or action it is later used to support.

5. Core Mechanism and Structural Sequence

5.1 Core Mechanism

source system is difficult to access, share, inspect, or reuse $\rightarrow$ synthetic or proxy representation is produced $\rightarrow$ representation preserves visible structure $\rightarrow$ fidelity metrics appear strong $\rightarrow$ downstream use is expanded $\rightarrow$ use requires relations not tested by fidelity metrics $\rightarrow$ hidden structural mismatch appears $\rightarrow$ representation becomes valid-looking but use-invalid

5.2 Structural Sequence

Failure Flow

representation generated → surface structure preserved → similarity appears sufficient → downstream use is inferred → use-specific structure is not checked → output supports an invalid inference or decision → fidelity is mistaken for validity

Aligned Flow

representation generated → preserved structure is specified → intended use is declared → use-required relations are identified → fidelity is tested against those relations → downstream validity is bounded → representation is used only within verified scope

6. Structural Components

source system synthetic representation proxy representation surface structure preserved distribution visible similarity dependency structure ordering structure causal or functional sequence downstream use use-validity evaluation metric validation scope decision consequence mechanism preservation fidelity-use gap

7. Primary Failure Patterns

1. Fidelity Overextension

representation matches source appearance → representation treated as valid for broader use

A narrow fidelity claim expands into a broader validity claim.

2. Surface-Structure Substitution

visible structure preserved → hidden required relation ignored

The representation carries the observable structure but not the relation required for the task.

3. Metric Misplacement

fidelity metric passes → downstream validity assumed

A metric designed for resemblance is used as if it validated use.

4. Ordered-Relation Loss

variables or components generated together → sequence or dependency relation is flattened

The representation may preserve co-occurrence while losing the order that gives the structure its use value.

5. Proxy-Use Expansion

proxy becomes available → proxy becomes operational substitute

A proxy created for access, privacy, exploration, or testing begins to replace the source in decision-relevant contexts.

6. Use-Invalid Realism

representation looks realistic → invalid inference becomes credible

The failure is strengthened, not weakened, by realism.

8. Operational Evaluation

1. What structure does the representation preserve?
2. What structure does the intended use require?
3. Are the preserved structure and the required structure the same?
4. Is the representation being evaluated only by surface similarity?

5. Does the downstream task depend on ordering, dependency, mechanism, causality, role sequence, or conditional relation?
 6. Has the representation been tested on the actual downstream use?
 7. Are fidelity metrics being used as substitutes for use-validity metrics?
 8. Does the representation preserve averages while losing subgroup, individual, temporal, or mechanism-level relations?
 9. Is the output being treated as evidence because it looks like the source?
 10. What would fail if the representation were structurally similar but use-invalid?
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9. Application Domains

1. Synthetic and proxy data

A synthetic dataset may reproduce general distributions while failing to preserve subgroup, causal, longitudinal, or record-level relations needed for the proposed analysis.

2. Simulation and scenario systems

Plausible scenarios may be useful for exploration but insufficient for intervention comparison if the mechanism connecting action and outcome is not preserved.

3. Summaries and compressed representations

A concise representation may retain topic structure while losing source conditions, exceptions, or sequence needed for decision or citation.

4. Evaluation and benchmarking

A fidelity score can become misplaced when it measures resemblance but is treated as evidence of utility, fairness, causal validity, or policy fitness.

5. Privacy-preserving transformation

A transformed object may improve access or confidentiality while changing the dependencies required for the intended research or governance use.

10. Relations to Adjacent Public Frameworks

1. Provenance-Validity Separation Model

Provenance–Validity Separation Model separates traceable origin from the validity, adequacy, or warranted use of an output. The present entry extends that distinction into a more specific interface, research, or responsibility condition.

2. Proxy Substitution as a Structural Governance Failure

Proxy Substitution as a Structural Governance Failure examines how a substitute can become operationally dominant while the original referent, participant, source, or responsibility relation disappears. The present entry specifies the substitution pressure at issue.

3. False Legibility Model

False Legibility Model describes how readable surfaces can be accepted before their underlying structure is understood. The present entry identifies a concrete way in which fluency, familiarity, or interface clarity can conceal an unresolved boundary.

4. Observer-Representation Boundary

Observer–Representation Boundary separates what an observer can read from what a representation can legitimately establish. The present entry specifies the evidential or operational consequences of crossing that boundary.

5. Boundary Failure Diagnostics

Boundary Failure Diagnostics provides a general vocabulary for locating non-entry, closure, lock-in, and boundary loss. The present entry narrows that diagnostic field to a repeatable mechanism.

11. Boundary Conditions and Non-Applicability

11.1 Boundary Conditions

This note applies when a representation is evaluated as faithful at one level but later used for a task that requires a different or deeper level of preservation.

The boundary is crossed when:

fidelity evidence → treated as use evidence

or:

similarity to source → treated as adequacy for downstream inference

It does not apply merely because a system produces a representation, proxy, summary, synthetic dataset, simulation, or model.

It applies when the representation's apparent fidelity exceeds the validated scope of its use.

11.2 Non-Applicability

This note should not be used to reject all synthetic representations, simulations, summaries, models, proxies, or generated data.

It does not apply well when:

the representation is used only for low-consequence exploration

the use case requires only surface-level resemblance

the required structure has been explicitly tested

the representation is not used for inference, decision, comparison, prediction, or intervention

the fidelity metric and the use-validity metric are properly separated

the representation's limits are clearly declared

It should not be reduced to:

synthetic representation is invalid

or:

proxy data cannot be useful

The narrower concern is:

a representation becomes use-invalid when its preserved structure does not match the structure required by the task.

12. Public Orientation and Method Boundary

This document is designed as a public conceptual entry. It provides a definition, mechanism, diagnostic orientation, and limit statement. It omits non-public editorial records, private source materials, non-public working materials, and implementation details.

The entry is not a substitute for empirical research, legal analysis, professional judgment, technical testing, or domain-specific governance. It should not be used to score people or institutions, automate high-stakes decisions, or infer misconduct from conceptual similarity.

Examples and application domains are illustrative. A case should be classified under **Structural Fidelity / Use-Validity Boundary** only when the relevant structure can be demonstrated from available evidence.

13. Limitations

1. Use-specific rather than universal

No representation is simply valid or invalid in the abstract. The assessment requires a declared task and consequence.

2. Metric dependence

The framework does not supply a universal fidelity metric or utility score.

3. Causal claims require separate methods

The note can identify a missing causal structure but does not establish causality.

4. No rejection of approximation

Approximate representations may be entirely adequate for bounded exploratory purposes.

5. Validation remains empirical

Conceptual separation cannot replace testing the representation on the actual downstream use.

14. Minimal Formulation

structural fidelity $\neq$ use-validity

distributional resemblance $\neq$ inferential adequacy

surface preservation $\neq$ downstream validity

synthetic similarity $\neq$ decision support

proxy realism $\neq$ use fitness

visible structure $\neq$ required structure

data-like output $\neq$ valid evidence

statistical fidelity $\neq$ causal adequacy

correlation preservation $\neq$ mechanism preservation

joint resemblance $\neq$ ordered dependency preservation

realistic representation $\neq$ valid intervention comparison

access restoration $\neq$ evidential sufficiency

output fidelity must be evaluated against intended use

a representation can look correct at one layer and fail at the layer where it is used

Looking structurally similar is not the same as being valid for use.

Alternative compressed form:

A representation may preserve appearance while losing the relation needed for action.

15. Conclusion

Structural resemblance and use-validity are related but non-identical properties. A representation becomes use-valid only when the structure it preserves is sufficient for the task it is asked to support. The more consequential the use, the less defensible

it is to infer adequacy from appearance, distributional similarity, or general realism alone.

The boundary established here is intentionally narrow. It does not prohibit representation. It requires that fidelity claims remain attached to the layer tested and that downstream use be validated against the relation the task actually needs.

Naming Declaration

Structural Fidelity / Use-Validity Boundary is a Meta-Writing Ecology term for the boundary between a representation's apparent structural similarity to a source and its validated adequacy for a specific downstream use.

The term structural fidelity refers to the preservation of visible, measurable, or recognizable features of a source system. These may include distributions, categories, proportions, correlations, sequences, output forms, layout, syntax, labels, or surface relations.

The term use-validity refers to whether the representation can legitimately support the task for which it is used. This may include inference, decision-making, comparison, prediction, intervention analysis, policy formation, evaluation, publication, citation, or operational action.

This term does not refer to any specific discipline, technical method, medical domain, dataset type, model provider, generative architecture, or empirical field.

Within Meta-Writing Ecology, the term functions as a Cross-supporting Note. It clarifies a recurring failure in which resemblance, accessibility, or synthetic realism is mistaken for evidential adequacy.

It should be used when a representation appears faithful to a source system but the downstream use requires structural relations that have not been preserved, tested, or declared.